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Procedure Quick-Reference

ORBIT NORTH · Week 3 operator pocket guide

ORBIT NORTH
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Faculty of Engineering · School of Engineering Design and Teaching Innovation (SEDTI)

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Procedure Quick-Reference

Twenty-five procedures students will most likely use during Week 3, in one-card format. The full procedure files are the authoritative source — these cards are pointers, not substitutes.

Card format: **ID · Title · Owner positions · Trigger · Top 3 steps · Exit criterion · File**.

Source of truth for procedures: `~/SpaceMissionSimulation/configs/eosat1/procedures/procedure_index.yaml` plus the per-procedure files in subfolders.

LEOP procedures

[Claude Code: produce a one-page card per procedure. Use this template:]

> **LEOP-001 · First Acquisition of Signal & OBC Boot** *Owner positions:* Flight Director · TT&C · FDIR/Systems *Trigger:* First pass over Iqaluit after separation timer expiry. *Top 3 steps:* (1) configure ground station, monitor RF acquisition; (2) verify bootloader beacon (SID 11); (3) issue OBC_BOOT_APP command (S8 func_id 55). *Exit:* App SW running, full HK reporting. *File:* `procedures/leop/first_acquisition.md`

Cards required for this section: - LEOP-001 First Acquisition of Signal & OBC Boot - LEOP-002 Initial Health Check - LEOP-006 Time Synchronisation - LEOP-007 Sequential Power-On

[Claude Code: also include LEOP-003 (Initial Orbit Determination) and LEOP-005 (Sun Acquisition) for advanced students who reach them in the simulation.]

Commissioning procedures

[Claude Code: one card per procedure listed below, same format as LEOP cards.]

- COM-001 EPS Checkout
- COM-002 TCS Verification
- COM-005 AOCS Mode Transitions
- COM-006 TTC Link Verification
- COM-007 OBDH Checkout
- COM-008 FDIR Configuration
- COM-009 Payload Power On
- COM-010 FPA Cooler Activation
- COM-011 Payload Calibration
- COM-012 First Light

Nominal procedures

[Claude Code: one card per procedure.]

- NOM-001 Pass Startup
- NOM-002 Imaging Session

- NOM-003 Data Downlink
- NOM-009 Routine Health Check
- NOM-012 Shift Handover

Contingency procedures (the most likely ones for Week 3)

[Claude Code: one card per procedure.]

- CTG-001 Under-Voltage Load Shed
- CTG-002 AOCS Anomaly Recovery
- CTG-003 TTC Link Loss Recovery
- CTG-007 Reaction Wheel Anomaly
- CTG-012 Overcurrent Response
- CTG-019 No Telemetry at Pass Start

Emergency procedures (read once, never expected to invoke)

Brief mention only — students should know these exist:

- EMG Emergency Safe Mode (`procedures/emergency/emergency_safe_mode.md`)
- EMG Loss of Attitude
- EMG Loss of Communication
- EMG OBC Reboot
- EMG Thermal Runaway

If you find yourself reaching for any of these, **call the FD first**. These procedures have safe-mode entry as a step and you do not invoke safe-mode without FD authorisation.

How to read a procedure file

[Claude Code: 200 words. The procedure files in the simulator follow a consistent structure. Walk through it using `procedures/leop/first_acquisition.md` as the example: front-matter, preconditions, steps with ownership column, verifications, exit criteria, troubleshooting.]

How procedures map to MCS commands

Each procedure step that involves commanding lists the PUS service to use (from `mcs/pus_services.yaml`):

Service	Use
S1	Telecommand verification (success / failure of last command)
S3	Housekeeping configuration (which SIDs report when)
S5	Event reports (anomaly events from the spacecraft)
S8	Function management (calling spacecraft functions, e.g. OBC_BOOT_APP)

Service	Use
S9	Time management (set time, time correlation)
S11	Onboard schedule (time-tagged commands)
S15	On-board storage and retrieval (downlink mass memory)
S17	Connection test (uplink-downlink link check)
S19	Event-action service (trigger an OBC action when an event fires)
S20	Parameter management (read / write OBC parameters)

You command via the MCS UI; you do not need to write PUS frames yourself.

When you are stuck

1. Pause. Use a callout: "[Position], standby."
2. Do not command. Re-read the procedure exit criterion.
3. Ask the FD for a 60-second hold.
4. If the FD agrees, log the hold and consult the manual section for the affected subsystem (`configs/eosat1/manual/0*.md`).
5. If you cannot resolve, the FD calls *no-go* and the team rolls back to the last verified state.

References

- Simulator procedure index: `~/SpaceMissionSimulation/configs/eosat1/procedures/procedure_index.yaml`
- ECSS-E-ST-70-32C — Test and operations procedures
- CCSDS Mission Operations Services Concept (CCSDS 520.0-G) — https://public.ccsds.org/Pubs/520x0g4.pdf
- NASA *Space Flight Program and Project Management Requirements* (NPR 7120.5F)